PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE

GOLD MUD PRESSURE FILTER POST

金泥压滤岗位

MTI-CHP-CPM-SOP-018_Rev. 000

Information Security Classification
B – Business Information

Rev.	Revision Status	Date	Prepared by	Reviewed by	Approved by
0	Issued for Use	24-APR-2024	*Approval as attached*	*Approval as attached*	*Approval as attached*
C0	Issued for Approval	24-APR-2024	Song Yuhua	Li Jianjun	Clark Wu
B0	Issued for Review	24-APR-2024	Song Yuhua	Li Jianjun	
A0	Issued for Internal Discipline Review	24-APR-2024	Song Yuhua		

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

DOCUMENT CHANGE STATUS

No.	Page	Section	Change Log	Revision by	Revision date	Revision version
1.						
2.						
3.						

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

Table of Contents

DOCUMENT CHANGE STATUS 2
Table of Contents 3
1. Purpose 目的 4
2. Functional Description 功能说明 4
3. References 参考 4
4. PPE 个人保护装备 4
5. Hazards & Control 危害与控制 5
6. Equipment 工具/ 设备 5
7. Procedure 程序 6
8. Gold Mud Pressure Filter Post 金泥压滤岗位 Questionnaire 18
9. Appendix 19

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the gold mud pressure filter post under complete closure.

此程序涵盖操作员对完全关闭下的金泥压滤岗位进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the gold mud pressure filter post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the gold mud pressure filter post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对金泥压滤岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定金泥压滤岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures.MTI-CP-SOP岗位操作规程
- Gold Mud Slurry Tank(4402-TNK-022).金泥储浆槽(4402-TNK-022)
- Gold Mud Slurry Tank Pump(4402-PSL-011/012).金泥渣浆泵(4402-PSL-011/012)
- Gold Mud Filter Press(4402-FIL-012).金泥压滤机(4402-FIL-012)
- Washing Water Tank (4402-TNK-018).洗涤水槽(4402-TNK-018)
- Washing Water Pump(4402-PSO-015).洗涤水泵(4402-PSO-015)
- Belt Conveyor(4402-BCV-005).皮带机(4402-BCV-005)
- Screwed Conveyor(4402-CNV-0014).螺旋输送机(4402-CNV-004)
- Gold Mud Ton Bag Packer(4402-BPR-004).金泥吨袋包装机(4402-BPR-004)

4. PPE 个人保护装备



This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Electric shock; 3. Slippery conditions and a trip to the danger; 4. Liquid corrosion; 5. Object falling from high altitudes; 6. Vehicle collision crush; 7. Liquid sputtering; 8. Noise. 1. 机械伤害; 2. 触电; 3. 湿滑的条件，绊倒危险; 4. 液体腐蚀; 5. 高空坠物; 6. 车辆碰撞碾压; 7. 液体喷溅; 8. 噪音。	1. Install a protective cover on the rotating part of the equipment; 2. Do not contact the electrical switch with wet hands during operation,Regular inspection of the equipment leakage protection device is complete, effective; 3. Please hold the handrail up and down the stairs; 4. Wear appropriate labor protection supplies, once the skin contact with hydrochloric acid solution, immediately rinse with a lot of water; 5. Wear labor protection goods and check the safety of stacking items on the operation platform; 6. Walk along the prescribed route, and stay away from the gold mud transport vehicles; 7. Wear protective masks and gloves, as well as other protective products as required; 8. Wear the earplugs to protect your hearing. 1. 设备的转动部位安装防护罩; 2. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效; 3. 上下楼梯请扶好扶手; 4. 穿戴适宜的劳保用品，一旦皮肤接触到盐酸溶液，立即用大量清水冲洗; 5. 穿戴好劳保护品，检查各平台的物品堆放安全性; 6. 按照规定的路线行走，远离金泥运输车辆; 7. 按要求戴好防护面具和防护手套，以及其它劳保护品; 8. 佩戴好耳塞，保护听力。

6. Equipment 工具/ 设备

4402-AGT-011	Agitator Of Gold Mud Slurry Tank金泥储浆槽搅拌器
	Clean the site tools清扫现场工具

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions. 1. 确认经过岗前技术培训; 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患; 3. 检查并打扫设备、现场卫生; 4. 与前后工序取得联系，并发出明确指令。
2	Inspect Gold Mud Slurry Tank (4402-TNK-022), Washing Water Tank (4402-TNK-018) 检查金泥储浆槽(4402-TNK-022)，洗涤水槽(4402-TNK-018)	1. Check whether the inlet and outlet pipelines and auxiliary pipelines, whether the installation of flange and valve meet the requirements, and whether the slurry storage tank leaks liquid; 2. Check the level control assembly is normal. 1. 检查出入口管线及附属管线，法兰、阀门安装是否符合要求，储浆槽是否漏液; 2. 检查液位控制组件是否正常。
3	Inspect Gold Mud Slurry Tank Pump(4402-PSL-011/012), Washing Water Pump(4402-PSO-015) 检查金泥渣浆泵(4402-PSL-011/012)，洗涤水泵(4402-PSO-015)	1. Check whether the pump inlet pipeline and auxiliary pipeline, flange and valve installation meet the requirements, whether the anchor bolts and ground wire are good, and whether the coupling is installed; 2. Turn the pump shaft 3 times, should be flexible without block; 3. Check whether the lubricating oil level is normal. The oil level should be between 1/2 and 2/3 of the oil lens; 4. Check the sealing condition of the pump is normal; 5. Point to move to confirm the correct steering; 6. Check whether the voltage meter, signal lamp and other meters of the pump control cabinet are normal; 7. Check whether the pump unit has air, if so, should be excluded; 8. Check whether the pump head is under the level of the slurry tank (4402-TNK-022). 1. 检查泵出入口管线及附属管线，法兰、阀门安装是否符合要求，地脚螺栓及地线是否良好，联轴器及防护罩是否装好; 2. 手转动泵轴3圈，应灵活无阻滞; 3. 检查润滑油油位是否正常，油位应在油镜的1/2 ~ 2/3

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		之间； 4. 检查泵的密封状况是否正常； 5. 点动确认转向正确； 6. 检查泵控制柜电压表、信号灯等仪表指示是否正常； 7. 检查泵机组是否有空气，如有应予以排除； 8. 检查泵头是否处于金泥储浆槽(4402-TNK-022)液位下。
4	Inspect Gold Mud Filter Press(4402-FIL-012) 检查金泥压滤机(4402-FIL-012)	1. Check whether the number of filter plates is sufficient, whether there is damage, whether the filter plate is clean, and whether the placement meets the requirements; 2. Check whether the filter cloth is folded, whether it is damaged, and whether the filter performance is good; 3. Check whether the pressure gauge of the mine pipeline is normal; 4. Check whether the connection is close and whether there is leakage; 5. Check whether the oil of the filter press is sufficient and whether the oil level meets the requirements (1 / 2~1 / 3); 6. Check whether the hydraulic oil top device is normal; 7. Check whether the emergency stop device is normal; 8. Check whether the pressure gauge of the hydraulic oil is normal and whether the pointer is in the specified interval; 9. Check whether the import and export of the filter press are smooth; 10. Automatic plate drawing device; 11. Whether the water connection plate is flexible and easy to use. 1. 检查滤板数量是否足够，有无破损，滤板是否清洁，安放是否符合要求； 2. 检查滤布是否折叠，有无破损，过滤性能是否良好； 3. 检查给矿管路压力表是否正常； 4. 检查各处联接是否紧密，有无泄露； 5. 检查压滤机油是否足够，油位是否符合要求(1/2 ~ 1/3)； 6. 检查液压油顶装置是否正常； 7. 检查急停装置是否正常； 8. 检查液压油的压力表是否正常，指针是否在规定的区间； 9. 检查压滤机各进出口是否畅通；

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		10. 检查自动拉板装置良好； 11. 接水盘是否灵活好用。
5	Inspect Belt Conveyor(4402-BCV-005) 检查皮带机(4402-BCV-005)	1. Each bearing is well lubrication and closed, the reducer oil level is appropriate, the upper and lower roller lubrication is good, the operation is flexible, whether the wear of the first and tail wheels and the tape are filled with materials, and whether the cleaning device of the first and tail wheels is in good condition; 2. Comprehensively check whether all parts of the belt conveyors are intact, whether the tension of the tape is appropriate, and whether the corresponding tape tightening device and dust-proof facilities are in good condition; 3. Whether the electrical facilities and control system are intact, whether the protective device is firm, and whether the funnel baffle and liner are damaged; 4. Check whether the cushion rollers of the impact load parts are intact; 5. Check the installation of belt sweeper; 6. Whether the installed electrical facilities and signal devices are intact and responsive; 7. Place the field control box in the manual position, press the start button, slowly rotate the potentiometer, and watch whether the running direction of the tape is correct, and whether the belt can run away; 8. After checking and contacting the post, press the start button; 9. During the operation process, check whether the tape machine emergency stop device, belt deviation correction and speed measuring sensor are in good condition. 1. 各部轴承润滑与封闭良好，减速机油位适当，上下托滚润滑良好，运转灵活，首、尾轮磨损情况与胶带间是否夹有物料，首、尾轮的清扫装置是否完好； 2. 全面检查皮带机各部分是否完好无损，胶带的张紧度是否合适，相应的胶带拉紧装置及防尘设施是否完好； 3. 电器设施和控制系统是否完好，防护装置是否牢固，漏斗挡板和衬板是否损坏； 4. 检查承受冲击载荷部位的缓冲托辊是否完好； 5. 检查皮带机清扫器安装是否正常； 6. 装设的电器设施及信号装置是否完好和反应灵敏； 7. 将现场控制盒置于手动位置，按一下启动按钮，迟缓

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		旋动电位器，观看胶带运转方向能否正确，皮带是否跑偏； 8. 检查无误并与前后岗位联系后，按下启动按钮； 9. 运转过程中，检查皮带机急停装置、皮带纠偏和测速传感器是否控制灵敏。
6	Inspect Screwed Conveyor (4402-CNV-004) 检查螺旋输送机(4402-CNV-004)	1. Check whether the lubricating grease in the bearing is appropriate; 2. Check whether the oil quality and quantity of the lubricating oil in the reducer are normal; 3. Check whether the bolts are fastened; 4. Check whether the cover plate of the screw conveyor must be tightly covered and whether the conveyor is clean; 5. Check whether the spiral blade is deformed ; 6. Determine whether the rotation direction of the motor meets the work requirements. 1. 检查轴承内润滑脂是否合适； 2. 检查减速机内的润滑油油质、油量是否正常； 3. 检查各螺栓是否紧固； 4. 检查螺旋输送机的盖板必须盖严，输送机内是否清理干净； 5. 检查螺旋叶片是否变形； 6. 判明电机旋转方向是否符合工作要求。
7	Inspect Gold Mud Ton Bag Packer(4402-BPR-004) 检查金泥吨袋包装机(4402-BPR-004)	a. Check whether the safety shield of each transmission part is intact and effective; b. Check whether the lubrication part has been grease; c. Check whether the indicator instrument is working normally, and the signal is sensitive and effective; d. Check whether the material scraping and fixing screw is loose; e. Check whether the pneumatic butterfly valve is normal and flexible; f. Check whether the lifting scale body platform is normal; g. Check whether the automatic uncoupling device is normal; h. Check whether the pneumatic clip is normal; i. Check the control system for normal operation. a. 检查各传动部位安全防护罩是否完好有效； b. 检查需要润滑部位是否已加油脂； c. 检查指示仪表工作是否正常，信号灵敏有效； d. 检查料刮固定螺丝是否松动； e. 检查气动蝶阀是否正常、灵活； f. 检查升降秤体平台是否正常；

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		g. 检查自动脱钩装置是否正常； h. 检查气动夹袋器是否正常； i. 检查控制系统是否正常。
8	Check the other Equipment and Facilities 检查其它设备设施	1. Agitator Of Gold Mud Slurry Tank(4402-AGT-011) a. Check whether the screws of the mixing paddle are fastened; b. Check whether the positive and reverse version of the agitator is correct; c. Check the surface of the mixer is smooth, without rough and pitting phenomenon; d. Check whether the mechanical seal of the mixer is good and whether the oil seal is normal; e. Electric mixer idling, check for abnormal sound, whether the up and down are concentric or shaking. 1. 金泥储浆槽搅拌器(4402-AGT-011) a. 检查搅拌桨的螺丝是否紧固； b. 检查搅拌器的正反转是否正确； c. 检查搅拌器的搅拌桨表面光滑，无粗糙和点蚀现象； d. 检查搅拌器的机械密封是否良好，各油封是否正常； e. 电动搅拌器进行空转，检查是否有异常声响，上下是否同心或摇晃。 2. Chute a. Check gold mud filter press (4402-FIL-012) whether the feeding chute is in good condition; b. Check gold mud filter press (4402-FIL-012) whether the blocking detection switch of the feeding chute is normal. 2. 溜槽 a. 检查金泥压滤机(4402-FIL-012)下料溜槽是否完好； b. 检查金泥压滤机(4402-FIL-012)下料溜槽的堵料检测开关是否正常。 3. Valve a. Check whether the butterfly valve (4306-PSL-001 / 002) of the gold slurry storage tank (4402-TNK-022) and the outer butterfly valve (100BF) at the bottom of the slurry storage tank (4402-TNK-022) are intact and

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		flexible; b. Check whether the three butterfly valves (125BF, 125BF, 125BF) at the entrance of the gold slurry pump (4402-PSL-011 / 012) and the two butterfly valves (100BF, 100BF) are intact and flexible; c. Check whether the inlet butterfly valve (100BF) of gold mud filter press (4402-FIL-012), compressed air ball valve (80BA), two washing water valves (100BF, 65BF) pumped by washing water pump (4402-PSO-015), and the filter discharge liquid butterfly valve (65BF) are intact and flexible d. Check whether a butterfly valve (50GA) in the washing water tank (4402-TNK- 018), and an outer butterfly valve (100BF) at the bottom , A valve (100BF) from the sponge copper filter (4402-FIL-001/002) is intact and flexible; e. Check whether the two butterfly valves (125BF, 125BF) at the inlet of the washing water pump (4402-PSO-015) and two butterfly valve (100BF,65BF) are intact and flexible； f. Check one ball valve (80BA) in the air pressure station (4601-CAT-001/002/003/004). 3. 阀门 a. 检查金泥储浆槽(4402-TNK-022)的金泥输送泵(4306-PSL-001/002)输送金泥管路蝶阀一个(125SP)，储浆槽(4402-TNK-022)底部的外排蝶阀一个(100BF)是否完好、灵活； b. 检查金泥渣浆泵(4402-PSL-011/012)的入口三个蝶阀(125BF,125BF,125BF),出口两个蝶阀(100BF,100BF)是否完好、灵活； c. 检查金泥压滤机(4402-FIL-012)入口蝶阀(100BF)，压缩空气球阀(80BA)，洗涤水泵(4402-PSO-015)泵过来的洗涤水两个蝶阀(100BF,65BF)，压滤机排液蝶阀(65BF)是否完好、灵活； d. 检查洗涤水槽(4402-TNK-018)的新水一个蝶阀(50GA)，底部的外排蝶阀一个(100BF)，来自海绵铜压滤机(4402-FIL-001/002)过来的滤液一个阀门(100BF)是否完好、灵活； e. 检查洗涤水泵(4402-PSO-015)的入口两个蝶阀

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		(125BF,125BF)，出口两个蝶阀(100BF,65BF)是否完好、灵活； f. 检查空压站(4601-CAT-001/002/003/004)的管路一个球阀(80BA)。
9	Inspect Materials Level 检查物料料位	1. Check whether the liquid level of the slurry storage tank (4402-TNK-022) and washing water tank (4402-TNK-018)meets the pressure and filtration requirements; 2. Check whether the material is stored in the gold mud filter press (4402-FIL-012), whether the hydraulic oil is sufficient, and whether there is a filter cake in the feeding chute； 3. Check whether the material is stored on the tape machine(4402-BCV-005), and there is no adhesive deposit on the surface of the head and tail wheel and the belt joint； 4. Check whether the material is stored in the screw conveyor (4402-CNV-004). 1. 检查储浆槽(4402-TNK-022)、洗涤水槽(4402-TNK-018)液位是否达到压滤要求； 2. 检查金泥压滤机(4402-FIL-012)内是否存料，液压油是否足够，下料溜槽是否存有滤饼； 3. 检查皮带机(4402-BCV-005)上是否存料，首尾轮表面与皮带结合部无黏附积矿存在； 4. 检查螺旋输送机(4402-CNV-004)内是否存料。
10	Start-Up Of The Gold Mud Pressure Filter Post 金泥压滤岗位开机	1. No-load start should be guaranteed when starting; 2. The driving sequence is along the material direction, from the back to drive that is, the reverse driving, taking into account to reduce the idling of the equipment, each equipment has been delayed to open; At the same time, Confirm that the process behind the gold mud pressure filter post has been started and has the feeding conditions: Open a new water butterfly valve (50GA) of the wash sink (4402-TNK-018) and close after filling with new water ⟹ Open a butterfly valve (125SP) from the mud pump (4306-PSL-001/002), The gold mud slurry storage tank (4402-TNK-022) is filled with gold mud mud mud ⟹ Start mud filter press (4402-FIL-012) Press the "start" button of the gold mud filter (4402-FIL-012), the oil pump starts to work, and then press the piston to press the pressure force reaches the high pressure, the

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		hydraulic system will automatically stop and maintain the pressure, open the butterfly valve of the filter press (100BF) and the liquid butterfly valve of the filter press (65BF)⟹Open the gold sludge slurry pump (4402-PSL-011 / 012) (100BF), start the gold sludge slurry pump (4402-PSL-011 / 012), open the inlet three butterfly valves (125BF, 125BF, 125BF)⟹The slurry is put into each filter chamber through the feed hole on the thrust plate, and then filtered under the feed pressure not greater than 0.6MPa to gradually form the filter cake ⟹After the filter cake is filled, close the inlet three butterfly valves (125BF, 125BF, 125BF) of the gold slurry pump (4402-PSL-011 / 012), stop the gold slurry pump (4402-PSL-011 / 012), close the gold slurry pump (4402-PSL-011 / 012) outlet two butterfly valves (100BF, 100BF) and the butterfly valve (100BF of the filter inlet (100 BF) ⟹Open the washing water pump (4402-PSO-015) outlet for two butterfly valves (100BF, PSO-65BF), Start the washing water pump (4402-PSO-015), Open the two butterfly valves (125BF, 125BF) at the inlet of the washing water pump (4402-PSO-015), to wash the filter cake at a pressure not greater than 0.6MPa. After the operation, close the two butterfly valves of the washing water (100BF, 65BF)⟹Slowly open the compressed air ball valve (80BA), dry the filter cake under a pressure no more than 0.6MPa, and close the compressed air ball valve (80BA) after operation⟹Press and hold the "Release" button on the operation panel, and the press plate will automatically stop after it is sent back in place, and the pull plate hook moves forward to unload the reciprocating plate⟹During the filter cake unloading process, Start the screw conveyor (4402-CNV-004)⟹ Start the gold mud ton bag packing machine (4402-BPR-004)⟹ Start the belt packing machine (4402-BCV-005) after the packing machine, and the forklift fork takes the ton bag. 1. 启动时应保证空载启动； 2. 开车顺序为沿物料走向，从后往前开即逆向开车，兼顾减少设备的空转，各设备相继延时开启；同时，确认金泥压滤岗位后面的工序已经开车并具备给料条件： 打开洗涤水槽(4402-TNK-018)的新水一个蝶阀(50GA)，并注满新水后关闭⟹打开金泥输送泵(4306-PSL-

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		001/002)过来的一个蝶阀(125SP)，待金泥储浆槽(4402-TNK-022)充满金泥泥浆⟶启动金泥压滤机(4402-FIL-012) 按下金泥压滤机(4402-FIL-012)“启动”按钮，油泵开始工作，然后再按一下“压紧”按钮，活塞推动压紧板压紧，当压紧力到达调定高点压力后，液压系统自动跳停并保持压力，打开压滤机入口蝶阀(100BF)和压滤机排液蝶阀(65BF)⟶打开金泥渣浆泵(4402-PSL-011/012)出口两个蝶阀(100BF,100BF)、启动金泥渣浆泵(4402-PSL-011/012)，打开入口三个蝶阀(125BF,125BF,125BF)⟶料浆通过止推板上的进料孔进去各个滤室，在进料压力不大于0.6MPa下加压过滤，逐渐形成滤饼⟶压滤机滤饼充满后，关闭金泥渣浆泵(4402-PSL-011/012)入口三个蝶阀(125BF,125BF,125BF)、停止金泥渣浆泵(4402-PSL-011/012)、关闭金泥渣浆泵(4402-PSL-011/012)出口两个蝶阀(100BF,100BF)和压滤机入口蝶阀(100BF)⟶打开洗涤水泵(4402-PSO-015)出口两个蝶阀(100BF,65BF)，启动洗涤水泵(4402-PSO-015)，打开洗涤水泵(4402-PSO-015)的入口两个蝶阀(125BF,125BF)，在压力不大于0.6MPa下对滤饼进行洗涤，操作完成后，关闭洗涤水两个蝶阀(100BF,65BF)⟶缓慢打开压缩空气球阀(80BA)，在压力不大于0.6MPa下对滤饼进行风干，操作完成后，关闭压缩空气球阀(80BA)⟶按住操作面板上的“松开”按钮，压板退回到位后会自动停止，同时拉板钩前进进行往复拉板卸料⟶滤饼卸料过程中，启动螺旋输送机(4402-CNV-004)⟶启动金泥吨袋包装机(4402-BPR-004)⟶ 吨袋包装机打包结束后启动皮带机(4402-BCV-005)，叉车叉走吨袋。
11	The Gold Mud Pressure Filter Post Stops 金泥压滤岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping; b. Stop in reverse order of driving order and material discharge order, Each equipment needs to stop: Close the gold slurry pump (4402-PSL-011/012) inlet

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		three butterfly valves (125BF, 125BF, 125BF), stop the gold slurry pump (4402-PSL-011/012), close the outlet two butterfly valves (100BF, 100BF)⟹ Close the inlet butterfly valve of the filter press (100BF), press the "stop" button of the gold mud filter press (4402-FIL-012), and turn off the power supply; Close the compressed air ball valve (80BA), Close the two butterfly valves (125BF, 125BF) at the inlet of the washing water pump (4402-PSO-015), Stop the washing water pump (4402-PSO-015), Close the wash water pump (4402-PSO-015) outlet of two butterfly valves (100BF, 65BF)⟹ Stop spiral Conveyor (4402-CNV-004)⟹ Stop the gold mud ton bag packing machine (4402-BPR-004)⟹ Stop belt conveyor (4402-BCV-005)⟹Clean the site for hygiene. 1. 正常停车 a. 停车时应待机内物料排净后再停车； b. 按照开车顺序逆序和物料排净顺序停车，各设备都需要延时停止： 关闭金泥渣浆泵(4402-PSL-011/012)入口三个蝶阀(125BF,125BF,125BF)、停止金泥渣浆泵(4402-PSL-011/012)，关闭出口两个蝶阀(100BF,100BF)⟹关闭压滤机入口蝶阀(100BF)，按下金泥压滤机(4402-FIL-012)“停止”按钮，关闭电源；关闭压缩空气球阀(80BA)，关闭洗涤水泵(4402-PSO-015)的入口两个蝶阀(125BF,125BF)，停止洗涤水泵(4402-PSO-015)，关闭洗涤水泵(4402-PSO-015)出口两个蝶阀(100BF,65BF)⟹停止螺旋输送机(4402-CNV-004)⟹停止金泥吨袋包装机(4402-BPR-004)⟹停止皮带机(4402-BCV-005)⟹打扫现场卫生。 2. Emergency Stop a. When an emergency occurs, Close spiral conveyor (4402-CNV-004), belt conveyor (4402-BCV-005), gold clay bag packer (4402-BPR-004), washing water pump (4402-PSO-015), gold mud slurry tank pump(4402-PSL-011/012) and gold mud filter press(4402-FIL-012), agitator of gold mud slurry

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		tank(4402-AGT-011), After all stop the operation, Check whether the start / stop button of each device is in the stop state, Close The gold mud delivery pump (4306-PSL-001/002) of the gold slurry storage tank (4402-TNK-022) delivers one butterfly valve (125SP), One external butterfly valve (100BF) at the bottom of slurry storage tank (4402-TNK-022), Three butterfly valves (125BF, 125BF, 125BF) (4402-PSL-011/012), two butterfly valves (100BF, 100BF), Gold filter (4402-FIL-012) (100BF), Two butterfly valves (125BF, 125BF) of the washing water pump (4402-PSO-015), two butterfly valves (100BF, 65BF), at the same time, the upper level of equipment all need to stop; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, the operation shall be restarted in the driving sequence; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，关闭螺旋输送机(4402-CNV-004)，皮带机(4402-BCV-005)，金泥吨袋包装机(4402-BPR-004)，洗涤水泵(4402-PSO-015)，金泥渣浆泵(4402-PSL-011/012)和金泥压滤机(4402-FIL-012)，金泥储浆槽搅拌器(4402-AGT-011)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭金泥储浆槽(4402-TNK-022)的金泥输送泵(4306-PSL-001/002)输送金泥管路蝶阀一个(125SP)，储浆槽(4402-TNK-022)底部的外排蝶阀一个(100BF)，金泥渣浆泵(4402-PSL-011/012)的入口三个蝶阀(125BF,125BF,125BF)、出口两个蝶阀(100BF,100BF)，金泥压滤机(4402-FIL-012)入口蝶阀(100BF)，洗涤水泵(4402-PSO-015)的入口两个蝶阀(125BF,125BF)，出口两个蝶阀(100BF,65BF)，同时，上一级设备均需停车； b. 事故处理后，设备恢复正常，如仍需要进行生产，按照开车顺序重新启动操作； c. 事故处理后，设备恢复正常，如不需要进行生

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		产，按照正常停车顺序进行操作。
12	Permits &Isolation Sheets 许可证和隔离表	1. Only authorized isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

8. Gold Mud Pressure Filter Post 金泥压滤岗位 Questionnaire

Declaration of Understanding

I (_____ Learner _____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____ Assessor _____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

9. Appendix

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

Standard Operating Procedure
标准操作程序

Gold mud pressure filter posts in the chloride plant
CP 金泥压滤岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
1					
2					
3					

Approvals 批准

Document Owner 程序持有人	Song Yuhua 宋裕华	Date: 日期: 2024.4.24	Signature 签字	
Processing Superintendent 氯化高级副经理	Cao Guibin 曹桂宾	Date: 日期: 2024.4.24	Signature 签字	
Processing Senior Manager 氯化高级经理	Li Jianjun 李建军	Date: 日期: 2024.04.24	Signature 签字	
Safety Department Manager 安全部经理	Airlangga Putra R.J	Date: 日期: 24.04.24	Signature 签字	
Deputy General Manager 副总经理	Li Xinhua 李新华	Date: 日期 2024.4.24	Signature 签字	
General Manager 总经理	Clark Wu 吴迎林	Date: 日期: 2024.4.24	Signature 签字	